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Deep learning for intracranial aneurysm segmentation using CT angiography

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Abstract

Objective. This study aimed to employ a two-stage deep learning method to accurately detect small aneurysms (4–10 mm in size) in computed tomography angiography images. **Approach.** This study included 956 patients from 6 hospitals and a public dataset obtained with 6 CT scanners from different manufacturers. The proposed method consists of two components: a lightweight and fast head region selection (HRS) algorithm and an adaptive 3D nnU-Net network, which is used as the main architecture for segmenting aneurysms. Segments generated by the deep neural network were compared with expert-generated manual segmentation results and assessed using Dice scores. **Main Results.** The area under the curve (AUC) exceeded 79% across all datasets. In particular, the precision and AUC reached 85.2% and 87.6%, respectively, on certain datasets. The experimental results demonstrated the promising performance of this approach, which reduced the inference time by more than 50% compared to direct inference without HRS. **Significance.** Compared with a model without HRS, the deep learning approach we developed can accurately segment aneurysms by automatically localizing brain regions and can accelerate aneurysm inference by more than 50%.

1. Introduction

An intracranial aneurysm (IA) is a saccular protrusion on the cerebral arterial vessel caused by a weak or damaged vascular wall. Intracranial aneurysms are the main cause of subarachnoid hemorrhage (Van Gijn *et al* 2007). Most aneurysms are 4–10 mm in size (Kassell and Torner 1983). It is difficult to detect small aneurysms clinically, and there have been cases of missed diagnosis. There are three main ways to treat IAs clinically: continuous observation, endovascular interventional embolization and surgical clipping (Ontario 2019). These treatments require medical imaging to detect abnormal regions early and provide more information for surgery. Due to the wide application of advanced imaging technology, such as computed tomography angiography (CTA) (Kouskouras *et al* 2004), an increasing number of IAs have been diagnosed. Real-time and accurate diagnosis is very important for the prognosis of patients with IAs. However, diagnosing IAs through medical imaging is time-consuming and laborious.

Vessel rendering technology has been shown to play a significant role in aneurysm diagnosis. Several studies have demonstrated that radiologists can identify more aneurysms through the interpretation of whole-brain virtual reality images than through the use of the maximum intensity projection method

(Tsuchiya *et al* 2001, Mallouhi *et al* 2003). Research conducted by Miki *et al* revealed that radiologists achieved better detection performance when presented with volume-rendered images (2021). However, vessel rendering technology is sensitive to noise and may provide insufficient contrast between vessels and surrounding tissues, potentially leading to false-positive results.

Deep learning has made great contributions to computer vision and medical image analysis (LeCun *et al* 2015). Deep learning neural networks efficiently process large amounts of medical image information to assist specialists in diagnosis (Litjens *et al* 2017). Lessmann *et al* (2019) proposed an iterative instance segmentation method based on a fully convolutional network (FCN) (Long *et al* 2015) for automatic vertebral segmentation in CT images. Based on an FCN, Ronneberger *et al* proposed U-Net (2015), which has become the most widely used network in medical image segmentation tasks. Many networks based on U-Net, such as U-Net++ (Zhou *et al* 2018) and nonlocal U-Nets (Wang *et al* 2020), have been proposed and have achieved better performance in medical imaging segmentation. Brosch *et al* (2016) applied a deep 3D convolutional coding network with shortcut connections to the brain Nuclear Magnetic Resonance Imaging segmentation of white matter lesions. Huang *et al* (2023) explored an automatic brain segmentation method based on 18F-fluorodeoxyglucose (18F-FDG) PET/MR dual-modality registration with an encoder-decoder architecture. Pang *et al* (2023) presented a novel adaptive weighted curvature-based active contour for ultrasonic and 3T/5T MR image segmentation problem. Recently, models (Gao *et al* 2021, Valanarasu *et al* 2021, Wang *et al* 2022) that adopt vision transformers and utilize attention layers to model global relations between tokens have been proposed. The spatial attention method proposed by Huang *et al* (2022) utilizes the advantages of images from different modalities. Wang *et al* (2022) proposed a multichannel attention network that combines information from various receptive fields. Shen *et al* (2023) designed a method based on an attention mechanism for the automatic outlining of subcutaneous fat. Zhao *et al* (2024) introduces a multimodal cross attention transformer network for Nasopharyngeal Carcinoma tumor segmentation using PET/CT images. These models show that deep learning is promising for medical image segmentation, and several deep learning-based automatic methods (Park *et al* 2019, Ueda *et al* 2019, Dai *et al* 2020, Jin *et al* 2020, Bo *et al* 2021) for detecting IAs via CTA, MRA, and DSA have been proposed. Zhao *et al* (2023) proposed a residual attention model for extracting multiscale features from MRA images. Yang *et al* (2022) used point-based 3D neural networks for aneurysm segmentation. Wang *et al* (2023) developed a method that uses the results from multiphase CTA to detect IAs. However, few current deep learning-based methods consider the segmentation of small aneurysms from multicenter datasets.

In this study, we explored a deep learning method for IA segmentation with CTA to improve segmentation effectiveness and accuracy. CT angiographic data from 956 patients were collected. These data were obtained from 6 institutions and a public dataset and were scanned by multiple devices. In the proposed approach, a lightweight and fast deep learning network is used to crop out the head regions of CTA images, and an adaptive 3D nnU-Net model is used as the main architecture to segment aneurysms. The network was trained on 100 CTA images and evaluated on 856 CTA images. We selected 100 cases for training to reduce the training time. The segmentation results generated by deep neural networks were compared with expert-generated manual segmentation results. Furthermore, the segmentation performance was assessed using Dice scores. Acute cerebrovascular events can be prevented by using improved AI algorithms to rapidly investigate aneurysms. These algorithms increase the detection rate for ultrasmall aneurysms, decrease the missed diagnosis rate, and increase the aneurysm detection rate.

Our method crops the dataset to the vicinity of the head region and implements nnU-Net (Isensee *et al* 2021) as the architecture for IA segmentation. This approach achieves high computational speed and accurate segmentation results for small objects. Moreover, the model provides the 3D relationship between the aneurysms and surrounding tissues for interventional embolization and surgical treatment.

The contributions of our method are as follows:

- We established a multicenter aneurysm dataset with images from 6 clinical hospitals (Guizhou Provincial People's Hospital, Tongren Municipal People's Hospital, Xingyi Municipal People's Hospital, The Second People's Hospital of Guiyang, People's Hospital of Anshun City and Zhijin People's Hospital), and the images were scanned by devices from different CT manufacturers (Siemens, Canon, Neusoft and United Imaging). The dataset will be available following an embargo from the date of publication.
- We developed a deep neural network approach for selecting brain regions, accelerating aneurysm inference and enabling the early detection of aneurysms before they grow and cause ruptures or subarachnoid hemorrhage.
- We implemented an adaptive segmentation algorithm and achieved accurate performance for aneurysms with an average diameter of 6.25 mm.

2. Methods

2.1. Data acquisition

We collected three internal datasets from 6 hospitals as well as one public dataset (the Large IA Segmentation Dataset¹⁰) in this study. As shown in figure 1(a), the internal dataset was obtained from 6 hospitals in Guizhou Province. In addition, we used 100 public data from GLIA-Net (Bo *et al* 2021) for training and the testing dataset for validation; this dataset is called external dataset G. CTA images from a total of 956 patients were used in this study. The detailed data distribution is shown in figure 1(d). Fifty out of 252 samples in external dataset G were devoid of aneurysms and served as negative controls to evaluate the model's precision (Van Grinsven *et al* 2016, Wang *et al* 2016, Litjens *et al* 2017). Patients at Guizhou Provincial People's Hospital were inspected with SOMATOM Definition Force and SIEMENS SOMATOM Definition AS+ equipment. A plain scan was routinely performed, a standard algorithm was used for image reconstruction, and the reconstructed slice thicknesses were 1.0 mm and 0.6 mm. After scanning, the original data were uploaded to a workstation, and 3D-CTA images were obtained through 3D reconstruction software processing. Other scan parameters for each hospital are shown in figure 1(a).

2.2. Annotation and splitting

The labeling work was carried out in three stages. In the first stage, a resident labeled the aneurysms with standard labels. In the second stage, a radiologist labeled the accuracy of the labeling from the first stage. Finally, all the labeling work was reviewed by a senior radiologist. The dataset was divided into a training set and a testing set. We selected 100 cases from external dataset G for training. The remaining 856 CTA images are used for model testing to assess the performance across multiple center datasets, which varies from 67 to 305 samples in each dataset. This choice helped us better understand the model's performance under different dataset sizes and avoid overfitting our model to one specific dataset (Campello *et al* 2021, Shanmugam *et al* 2021).

2.3. Data collection phase

All the 3D-CTA images were acquired in DICOM format. The size of each image is $512 \times 512 \times n$, where n ranges from 176 to 1314. The cross-sectional images are 512×512 in size and are superimposed layer by layer along the z -axis from the inferior to superior direction. Annotation files for the aneurysms were obtained from the physicians (figure 2). Since the average diameter of an aneurysm is approximately 5 mm, an aneurysm typically occupies approximately 10 slices of the volume. Hounsfield units (Schneider *et al* 1996) (HUs) make up the grayscale in medical CT imaging and range from -1024 HU (black) to 3071 HU (white).

2.4. Head region dataset acquisition phase

Each CTA image consists of 176–1314 image slices, which indicates that most patients underwent head and neck vascular scanning, while a few patients underwent head, neck and heart one-station scanning for the simultaneous diagnosis of heart disease. Redundant information in the neck and heart regions will increase the segmentation workload. However, the existing skull stripping methods still need the entire 3D CTA image as input, which requires a large amount of memory and computation. Therefore, we performed a head region selection (HRS) preprocessing step to exclude additional regions beyond the head region and to reduce the computational cost.

We trained a deep neural network called SegNet (Badrinarayanan *et al* 2017). It quickly accepts a pair of 2D images and crops the selected brain region of the patient. We manually annotate the head regions as the ground truths and select samples for HRS training. We preobtained 3D brain masks using a skull stripping algorithm (Bauer *et al* 2013). We did not use the head regions directly extracted from this algorithm because the algorithm is unstable, and approximately 20% of the predicted brain mask regions exceeded the real brain regions in our datasets. We selected stripping samples with complete head regions as the training set for the HRS phase. First, we clipped the original CTA image HU values to the interval of the bone window. Second, the bone window image and the results of the ITK-Skull stripping algorithm (Bauer *et al* 2013) were both maximally projected (Mroz *et al* 2000) in the sagittal and coronal planes. The sagittal and coronal plane images were used as inputs of the training set, and the skull stripped results were used as labels of the training set. Since both the inputs and labels were maximally projected, they were in 2D format.

2.5. HRS phase

We developed a feasible method in which head regions are cropped from CTA images without complex or time-consuming computations, and the process is shown in figure 2(c). After SegNet is trained on the

¹⁰ <https://doi.org/10.5281/zenodo.6801398>.

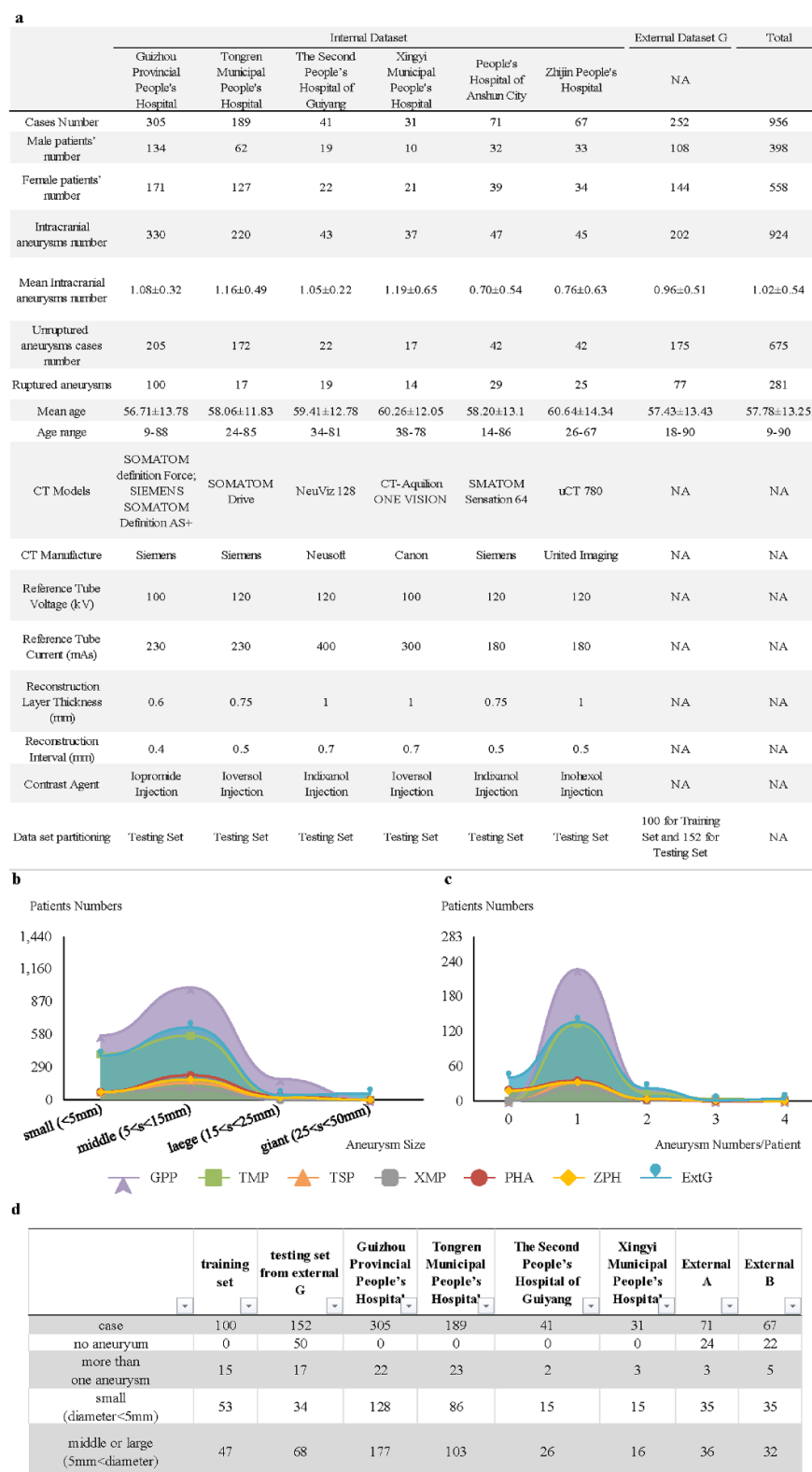


Figure 1. Patient and CTA characteristics of different hospitals. (a) Patient characteristics and CT machine information from the internal and external datasets. (b) Distributions of aneurysm size in different hospitals or datasets. The aneurysm size is calculated in terms of diameter, and aneurysms are grouped into four subsets, i.e. small (<5 mm), middle (5–15 mm), large (15–25 mm) and giant (25–50 mm) aneurysms (Ueda *et al* 2019). (c) Distributions of the number of aneurysms per patient. The abbreviations in the caption are the hospital names. GPP: Guizhou Provincial People's Hospital. TMP: Tongren Municipal People's Hospital. TSP: The Second People's Hospital of Guiyang. XMP: Xingyi Municipal People's Hospital. PHA: People's Hospital of Anshun City. ZPH: Zhijin People's Hospital. ExtG: External Dataset G. (d) Patient characteristics from the training set and testing set.

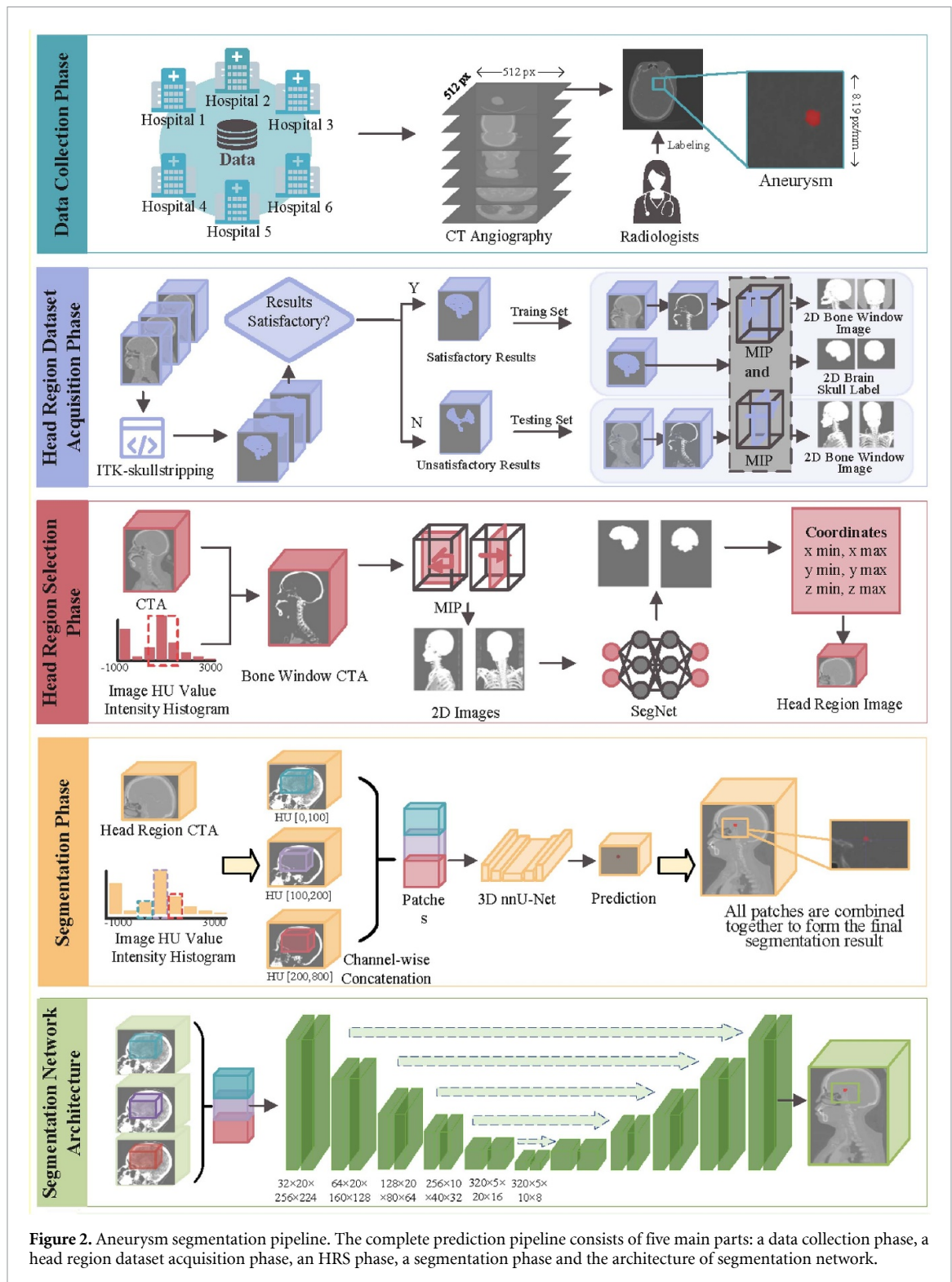


Figure 2. Aneurysm segmentation pipeline. The complete prediction pipeline consists of five main parts: a data collection phase, a head region dataset acquisition phase, an HRS phase, a segmentation phase and the architecture of segmentation network.

homemade dataset, the head region from each CTA image is cropped through the HRS process. Each image is thresholded using the bone range [200–800], and voxels outside this range are set to 0 HU. A bone window image is used because the maximum projection of the bone window clearly shows the outline of the human body. Bone window images are subjected to maximum projection along the x -axis and y -axis and sent to SegNet to generate two corresponding brain segmentation results. These two results are on the sagittal and coronal planes, through which the minimum and maximum approximate coordinates of the three directions are obtained. The brain region is cropped via 6 coordinates: the minimums and maximums of the x , y and z coordinates. The above HRS phase in the preprocessing stage involves rapid segmentation of the brain region. The coordinates of the brain region are used to crop the original CTA images into images containing only the head.

2.6. Segmentation phase

The processed head region dataset is clipped into three channels according to the HU scale values, where the values are within the ranges of [0, 100], [100, 200] and [200, 800]. The CTA images are clipped to [0,100], [100, 200] and [200, 800] HU windows to clearly display blood vessels, plasma, body tissues and bone (Mamourian 2013). Then, each channel image is normalized with the z score. The three images of different HU values are concatenated into one tensor along the channel dimension as the input of the segmentation network.

An adaptive 3D convolutional neural network (CNN) based on the nnU-Net architecture is implemented to provide accurate segmentation of IAs on head CT images. It is an adaptive U-Net-based framework that removes some flashy structures and retains only the basic structures. The nnU-Net automatically configures and trains three distinct network architectures: a 2D U-Net, 3D full-resolution U-Net, and 3D U-Net Cascade. The 2D U-Net is trained and inferred on axial slices, while the 3D U-Net is trained and inferred on the original full-resolution data. The 3D U-Net is the most powerful single-model structure, but its patch size is limited by GPU memory. The 3D U-Net Cascade initially trains a 3D U-Net at low resolution to generate a rough segmentation, followed by a full-resolution 3D U-Net to correct and enhance the details. This approach allows for the aggregation of more contextual information. The nnU-Net selects the optimal single model or model combination based on cross-validation results for prediction. The network architecture is shown in detail in figure 2(d). The size of the nnU-Net convolution kernel is $3 \times 3 \times 3$, and the pooling operations are performed 5 times, 5 times and 4 times on the z -axis, y -axis and x -axis, respectively. The feature maps obtained from the downsampling module are of dimensions $32 \times 50 \times 256 \times 224$, $64 \times 20 \times 160 \times 128$, $128 \times 20 \times 80 \times 64$, $256 \times 10 \times 40 \times 32$, and $320 \times 5 \times 20 \times 16$.

The batch size was set to 2, which is the maximum allowed by the GPU. The patch sizes used in our study were determined through an iterative process. First, the batch size was fixed at 2, and then a relatively large initial patch size was set. The network configuration was adjusted based on the current patch size. Subsequently, the patch size was continuously and iteratively adjusted based on the memory consumption. We selected [160, 112, 128] as the patch size following this method to balance the need to capture contextual information with the computational resource constraints. The area of an aneurysm is usually very small. To prevent sample imbalance during training, we sampled more than 1/3 of the pixels to foreground pixels in a batch. The initial learning rate for training was set to 0.002, and the weight decay was set to 3×10^{-5} .

3. Results

3.1. HRS performance

We performed the HRS step on all CTA images to reduce both training and testing times. We selected 100 cases to evaluate the time savings realized by using HRS and not using it. Processing 100 training data points without HRS required 1700 s per epoch, compared to only 800 s per epoch with HRS.

The HRS method localizes brain regions. Figure 3(a) shows a qualitative comparison between the ITK-Skull stripping method and the HRS method. The ITK-Skull stripping method extracts the brains from CTA images accurately in some instances and outrageously in other instances. The first column of figure 3(a) shows the brain regions extracted by this method, from which we observe that the predicted brain regions deviate from where they are supposed to be. In the second column of the figure, we observe that the brain regions predicted by the HRS method are consistent with the location of the brain, which enables the CTA images (the third column of figure 3(a)) to be divided into regions near the brain (the fourth column of figure 3(a)).

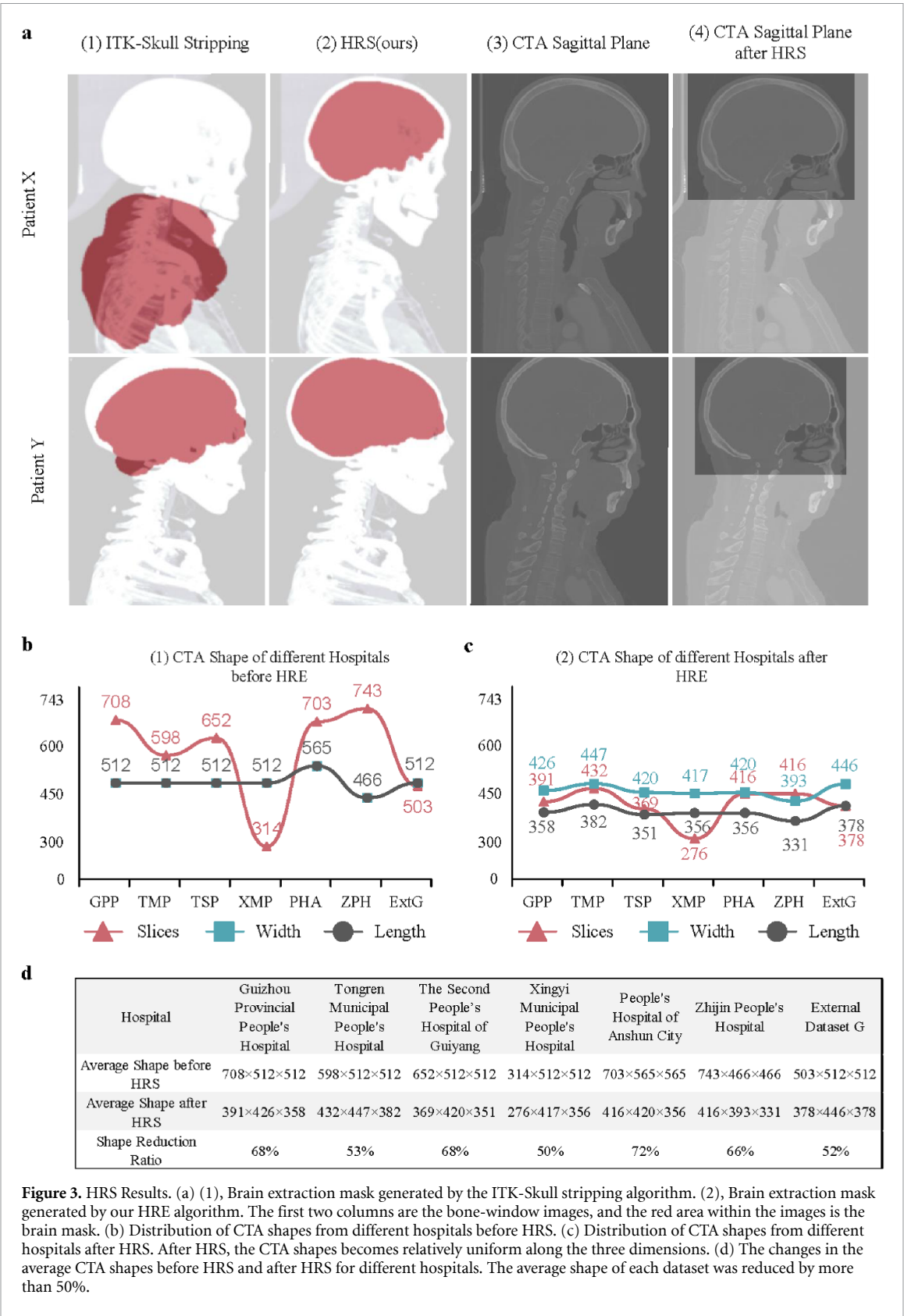
All the results produced by HRS are similar to those shown in the second column of figure 3(a). Since there is no accurate ground truth for the brain region, we do not use any segmentation performance metrics.

Figures 3(b)–(d) show comparisons of the shapes of CTA images from different hospitals before and after the HRE process. The average image size was reduced by more than 50% (figure 3(c)). In the inference stage, due to the large size of some original CTA images, nnU-Net was not able to infer the results in finite time with a GPU, but for the processed data, only a few seconds were needed to infer the results.

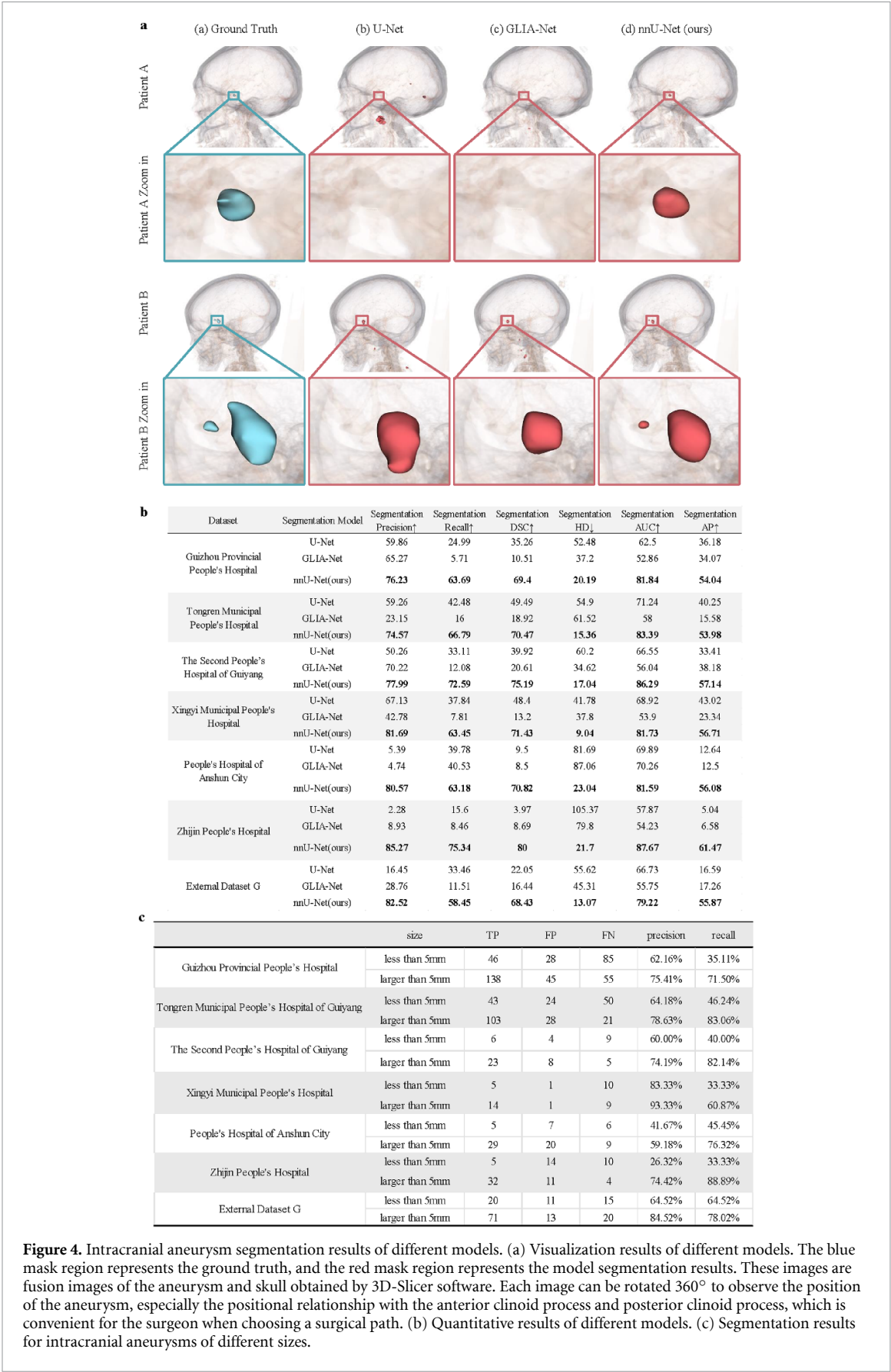
3.2. Segmentation performance

To better demonstrate the segmentation performance of our model, we carried out experiments using two other deep learning segmentation models, U-Net and GLIA-Net. These two models were trained on the same 100 training examples.

Figure 4 shows a qualitative comparison of the different methods. Both U-Net and GLIA-Net made false-positive predictions that some sinus venous veins were aneurysms. However, the nnU-Net model we



adopted better distinguished between aneurysms and venous sinuses. The aneurysm in Patient A was small, and U-Net and GLIA-Net failed to identify it, while nnU-Net successfully identified it. Patient B had two aneurysms. Both U-Net and GLIA-Net were able to identify the larger aneurysm but were not able to detect the smaller aneurysm. The nnU-Net model successfully identified both aneurysms. We also found that both U-Net and GLIA-Net segmented the main shape of the larger aneurysm in Patient B. However, the segmentation result of nnU-Net was meticulous for the finer parts at both ends of the aneurysm.



The quantitative results of aneurysm segmentation at the pixel level by the three models are shown in figure 4(b). The precision exceeded 74% on the different datasets. The recalls on the 6 hospital datasets and external dataset G were 63.69%, 66.79%, 72.59%, 63.45%, 63.18%, 75.34% and 58.45%, respectively.

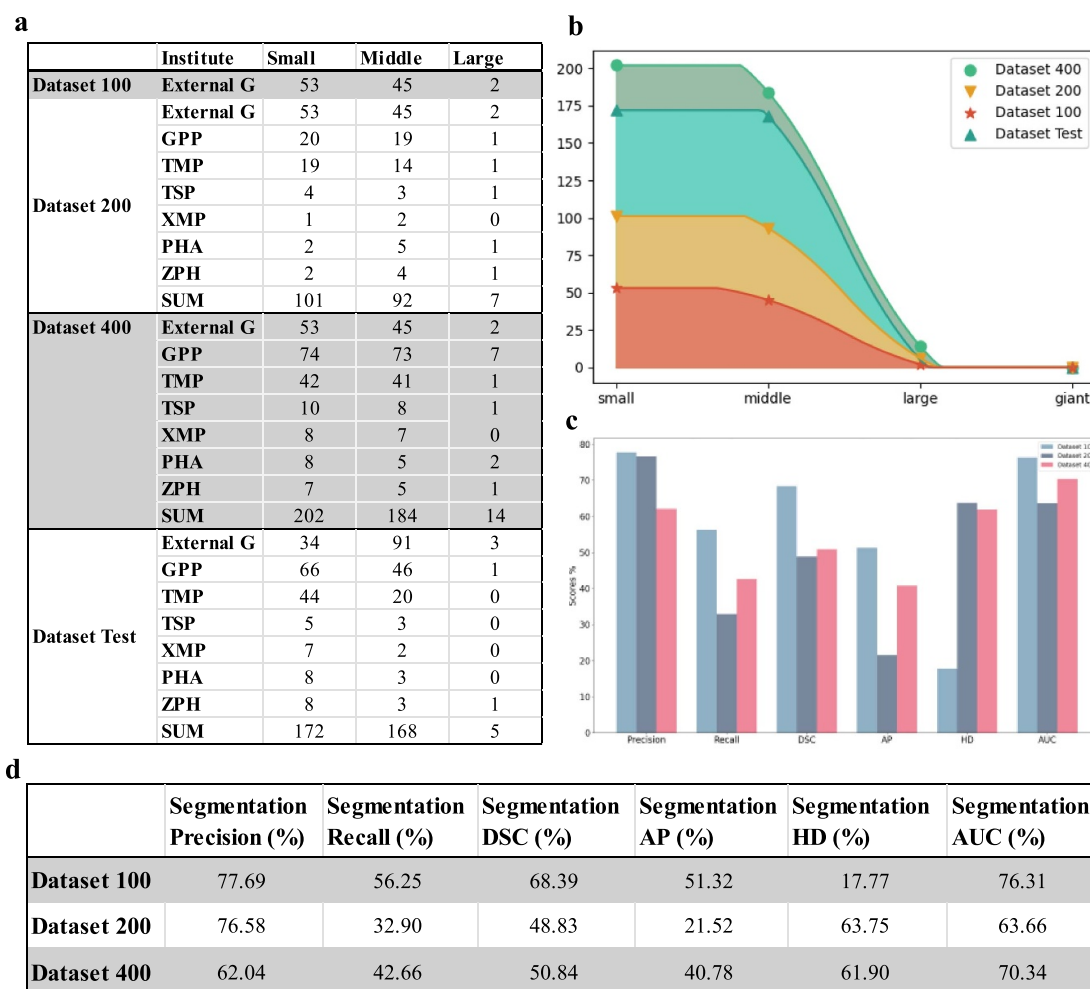


Figure 5. Experimental analysis using datasets consisting of 100, 200, and 400 samples. (a) Aneurysms numbers across training and testing sets. (b) Distribution of aneurysms scales from different hospitals (c) Quantitative results of models trained on datasets of various scales. Dataset 100, Dataset 200, and Dataset 400 represent the models' trained on 100, 200, and 400 CTA images, respectively. (d) Bar graph of the outcome metrics comparison among the three models.

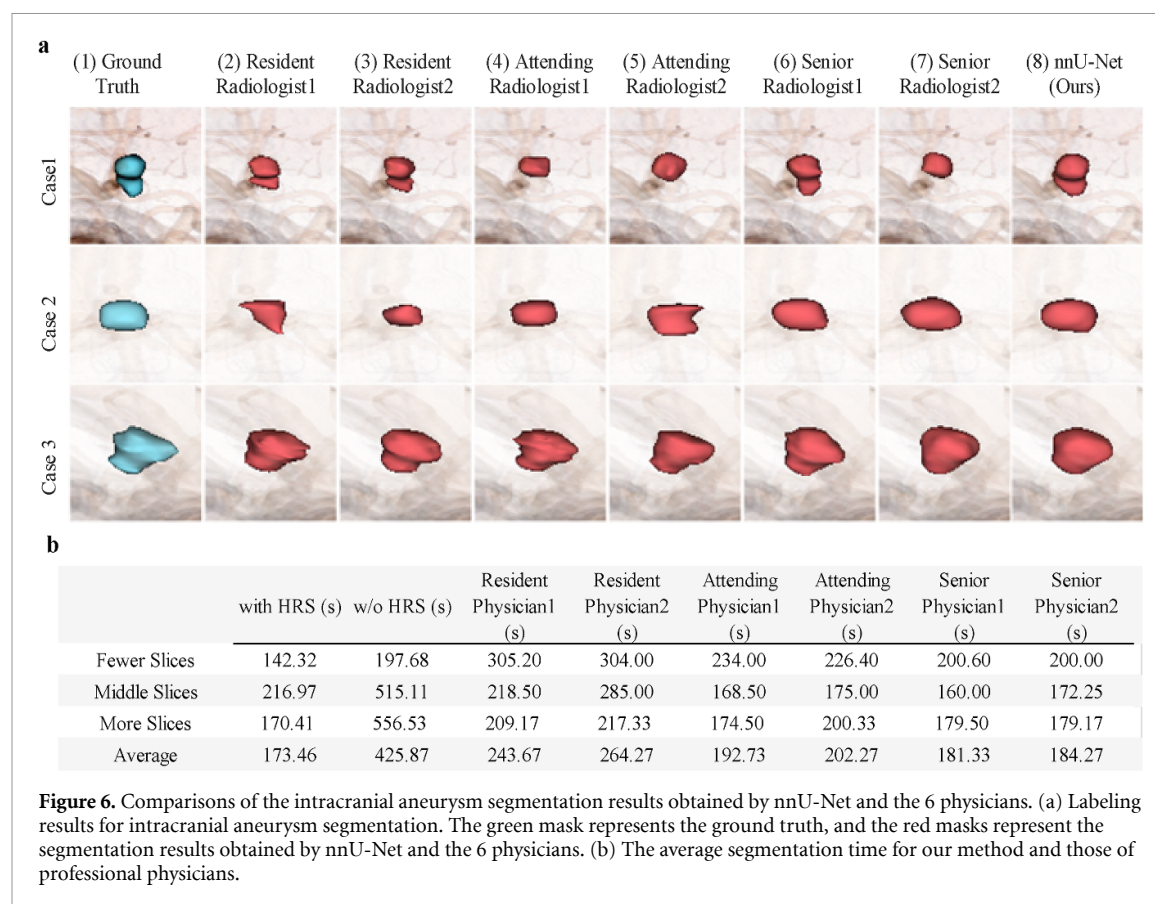
Moreover, we obtained statistical results at the tissue level. As shown in figure 4(c), for aneurysms whose diameters were larger than 5 mm, the recalls on the 6 hospital datasets and the external dataset G were 71.50%, 83.06%, 82.14%, 60.87%, 76.32%, 88.89% and 78.02%, respectively.

From these results, we found that the precision was generally higher than the recall, indicating that the method we used has stronger specificity, which is very important in the medical field. We also used the dice similarity coefficient (DSC) and the 95th percentile Hausdorff distance (95% HD) as indicators to measure the effect of 3D medical image segmentation.

We further compared the results obtained when the model was trained on datasets of different scales. The distribution of the new training and testing sets is listed in figures 5(a) and (b). Beyond the initial training and testing sets, we have selected additional images from other hospitals to expand the training datasets to 200 and 400 samples. Concurrently, a new multicenter test set has been established by selecting images from the remaining dataset. As seen in figure 5(b), the proportions of aneurysms of varying size across the three datasets are comparable. As shown in figures 5(c) and (d), the performance of the model trained with 100 CTA images was superior to those of the other two models on all six metrics (precision, recall, DSC, AP, HD and AUC). Specifically, the precision, recall, DSC, AP and AUC were greater than those of the model trained with 400 images (all above 6%). The performance of these three models proves that our proposed models' generalizability on multi-center dataset. In addition, the training time of the model with 100 CTA data points was far less than those of the other two models (3 d vs. 6 d and 8 d, excluding the validation time).

3.3. Segmentation performance comparison of the model and radiologists

We selected 15 thin-layer axial images of IAs of different sizes and excluded 3D reconstructed images for double-blind diagnosis. Six radiologists, including two resident radiologists, two attending radiologists and



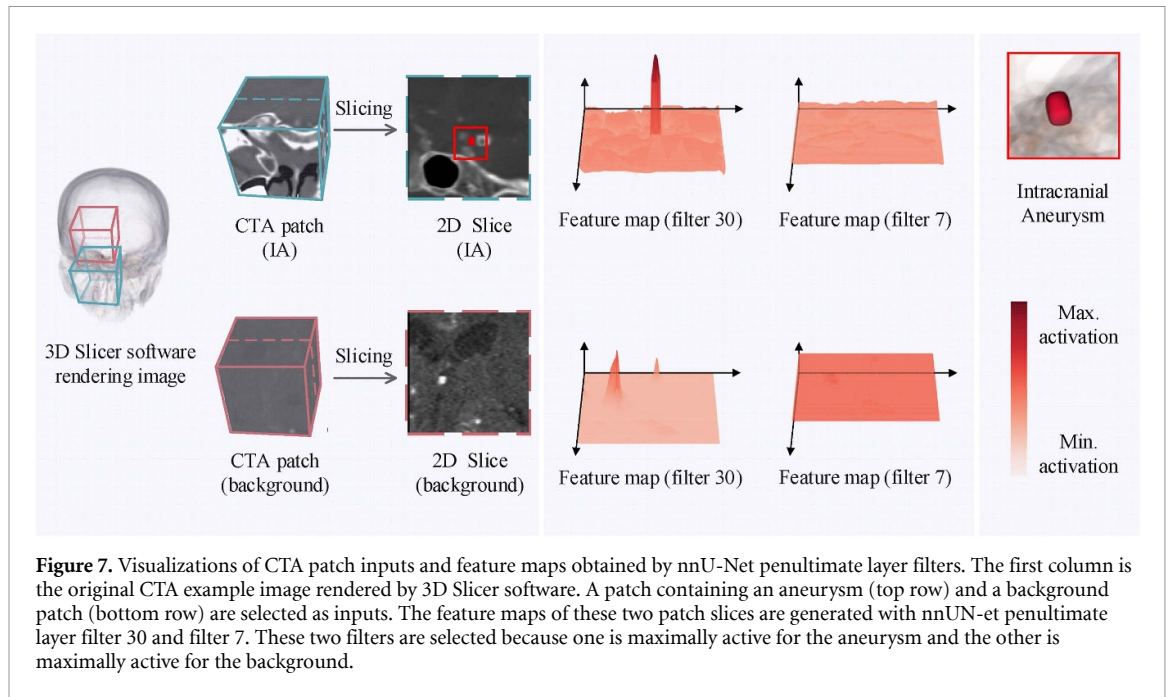
two senior radiologists, and our deep learning network model participated in the diagnosis process. By comparing the diagnoses from these multiple doctors, we aimed to validate the efficiency of our method in terms of diagnostic accuracy. The aneurysms labeled by the 6 physicians are shown in figure 6(a). The segmentation of aneurysms by the radiologists was divided into two steps: diagnosis and labeling. For fairness, we summed the diagnosis time and labeling time and compared this sum with the model segmentation time. If a radiologist failed to find the aneurysm after more than 300 s during the diagnosis process, both the diagnosis time and the labeling time were set to 300 s.

As shown in figure 6, both the CNN and the radiologists located the aneurysms accurately. Since the ground-truth segmentation results and the segmentation results generated by the radiologists were drawn with ITK-SNAP software, the edges of the aneurysms are coarse, while the aneurysm prediction edge segmented by nnU-Net is smoother. The aneurysm in Patient 3 was a fusiform aneurysm. The nnU-Net model segmented the main part of the aneurysm, which achieved the same segmentation result as primary radiologist 1.

The average segmentation times of nnU-Net with HRS, nnU-Net without HRS and the 6 radiologists are shown in figure 6(b). The deep learning network model without HRS took 425.87 s and reached 173.46 s with HRS on average. The deep learning network model with HRS required 31.59%, 12.12% and 5.10% less time than the two radiologists and 59.26% less time than the deep learning network model without HRS. Since there were only thin-layer axial images, the radiologists missed diagnoses in 3 patients, while our deep learning network model missed diagnoses in 2 patients.

4. Discussion

Our study describes a fully automated method for segmenting aneurysms based on deep learning. The system has two components. The first is a network called SegNet for HRS that segments brain regions from a pair of axial maximum projection images. The segmented brain images are then used to generate the coordinates of the cut head region. HRE only needs a pair of two-dimensional input images, and it quickly locates the brain area. HRS is faster than ITK-Skull stripping and faster than directly using a 3D image as input. The second network is called the nnU-Net 3D segmentation model, which finely segments IAs. This adaptive algorithm has high diagnostic accuracy for aneurysms.



The presence of vessel overlap is not inherently problematic for 3D imaging; however, we acknowledge the potential influence of intricate vessel morphology on the effectiveness of our algorithm. To address this concern, we adopted the strategy of clipping images with varying HU values during the data collection phase. This approach simplifies the complexity of vascular structures and enhances the performance of our algorithm. In future studies, we aim to further explore the impact of complex vessel morphology on the performance of our algorithm and to investigate methods for optimizing its performance in the context of intricate vessel structures.

In this study, small aneurysms were segmented in their early stages via a deep learning technique. This can serve as a helpful reminder to patients about the value of prevention, maintaining normal blood pressure, and routine follow-up to assess changes in aneurysm volume to prevent aneurysm hemorrhage caused by hypertension, which is crucial for preventing subarachnoid hemorrhage.

Doctors can clearly visualize the 3D interaction between an aneurysm and the anterior process using 3D-Slicer. They may also assess and direct the clamping technique and decide whether to remove the anterior process to replicate the surgical course. When an aneurysm is near the anterior process, either interventional embolization or resection of a piece of the anterior process should be used.

As figure 7 shows, nnUN-et penultimate layer filters have the ability to recognize aneurysms and the background. The overall activation value of the background patch is low for filter 30, indicating that filter 30 pays less attention to the background patch. The activation values of all regions are almost the same for filter 7, and the activation values of the background patch are greater than those of the patch containing an aneurysm. This indicates that filter 7 is able to recognize all brain regions except for the aneurysm and is more active on the background.

A deep 3D CNN is able to segment aneurysms well and satisfies the clinical requirements for accurate and standardized analysis. In previous works, several traditional methods were employed for the segmentation of aneurysms. Hernandez *et al* (2007) distorted aneurysm contours by minimizing the energy function. Hentschke *et al* (2011) used the k-means algorithm to perform clustering and then detected aneurysms. Hemalatha *et al* (2018) modified the active contour model method. However, these traditional methods rely on physicians manually selecting the regions of interest or seed points of an image, which requires laborious manual interaction operations, and the segmentation outcome still needs to be improved. Recently, several automatic aneurysm segmentation methods based on deep learning have been proposed. HeadXNet (Park *et al* 2019) uses an encoder-decoder structure. The encoder is adapted from a 50-layer SE-ResNeXt model (Hu *et al* 2020) and pretrained on the Kinetics-600 dataset, and the decoder is a sequence of 3×3 transposed convolutions. Aneurysm datasets are usually large, and the addition of extra datasets to HeadXNet for pretraining inevitably increases the duration of the entire process. GLIA-Net (Bo *et al* 2021) uses a global positioning network. The global localization-based IA network integrates global localization priors to generate fine-grained 3D segmentation results. However, GLIA-Net obtains global information throughout the whole CTA image and takes hours to train one epoch.

Our study has several limitations. First, the model has a certain degree of underdiagnosis, but it focuses mainly on small aneurysms with diameters of approximately 5 mm. The underdiagnosis rate for aneurysms with diameters smaller than 5 mm and those larger than 5 mm are 58.73% and 23.08%, respectively. The average small aneurysms pixel occupies only 0.0142% of whole input patch, which poses a challenge to their detection. As depicted in figure 4(c), we have calculated the underdiagnosis rate of small aneurysms from different vendors, which are 64.89%, 53.76%, 60%, 66.67%, 54.55%, 66.67% and 35.48%, respectively. However, the numbers of small aneurysms from different hospitals exhibits considerable variation (ranging from 11 to 131). In future work, we plan to augment the dataset and assess the generalization capability of our model. Second, although our method reduces the computing time, it requires approximately 6 GB of GPU memory to compute the 3D segmentation in the inference phase. In the future, we will focus on the lightweight design of the model to facilitate its deployment on a general-purpose machine. Third, information on how an aneurysm changes in size and shape over time can be valuable for predicting the medical condition of the patient, and the current work has focused solely on the volume and shape of the aneurysm in the current scenario. Fourth, there is no related information about the beam, motion blur or beam hardening in current CTA images. In future studies, we plan to collect additional data, such as images with artifacts, and conduct further experiments to assess the performance of our method under these circumstances. In the future, we will also investigate determining the risk of aneurysm growth or rupture.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary information files).

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